

# Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of  $\mathcal{ALCI}_{\text{RCC5}}$  — a machine-checked campaign

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with AI assistants: Claude Opus 4.6–4.8, Fable 5, Opus 5 (Anthropic);  
GPT-5.4, 5.4 Pro, 5.5, 5.6 Pro, 5.6 Sol (OpenAI)

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# The one-slide version

- In 2002/2003 I defined a family of **spatial description logics** and left their central decidability question **open**. It still is.
- In 2026 I directed a **five-month AI campaign** on it: frontier models, adversarial review, machine verification.
- The **full** question is **not settled** — but it is *reduced* to **one well-posed lemma**, with an exact arithmetical address ( $\Pi_1^0$ ) on the way.
- A **sub-question is settled**: drop  $\forall$ PO and the logic is **decidable**, by a procedure the **Lean kernel checks**.

## How to read everything in this talk

The mathematics is AI-produced and *unverified by human experts*. Standing label for the **full logic**: **strongly supported**, **not certified** — but the  $\forall$ PO-free fragment is **machine-certified**, reviewed three times.

## Five ways two regions can relate (RCC5)



DR

disjoint



PO

partial overlap



EQ

equal



PP

proper part



PPI

proper superpart

- Every pair of regions stands in **exactly one** of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any “what contains / touches what” domain — **no coordinates required**.

## Reasoning without geometry: the composition table

If I know  $x R y$  and  $y S z$ , what can  $x ? z$  be?

| known    | known    | forced / allowed                                      |
|----------|----------|-------------------------------------------------------|
| $x PP y$ | $y PP z$ | $x PP z$ ( <i>forced</i> : parts of parts are parts)  |
| $x PP y$ | $y DR z$ | $x DR z$ ( <i>forced</i> : inside something disjoint) |
| $x DR y$ | $y DR z$ | anything at all ( <i>free</i> : 5 options)            |

- All 25 cells form the **composition table** — pure algebra, derived once from set semantics, then used forever.
- **Abstract semantics**: a “model” is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

## A description logic: you write definitions and facts

A **TBox** is a set of concept *definitions*, built from properties and **quantified spatial relations**:

$\text{CityAtRiver} \equiv \text{City} \sqcap \exists \text{PO}. \text{River}$       a city *overlapping* a river

$\text{GermanCity} \equiv \text{City} \sqcap \forall \text{PP}. (\neg \text{Country} \sqcup \text{Germany})$       a city in *no country but* Germany

Then we **assert** a few primitive facts — and nothing more:

$\text{Hamburg} \sqsubseteq \text{City} \sqcap \exists \text{PO}. \text{Alster}$       (a city, overlapping the Alster)

$\text{Alster} \sqsubseteq \text{River} \sqcap \exists \text{PP}. \text{Germany} \sqcap \forall \text{PO}. \neg \text{Country} \sqcap \forall \text{PP}. (\neg \text{Country} \sqcup \text{Germany})$

Note what we did *not* say: that Hamburg is a GermanCity, or a CityAtRiver.

## ... and the reasoner *infers* the classification

We never *asserted* that Hamburg is a GermanCity or a CityAtRiver — the **reasoner derives both**.

**City at a river** (immediate):

Hamburg PO Alster  $\wedge$  Alster  $\sqsubseteq$  River  $\implies$  Hamburg  $\sqsubseteq$  City  $\sqcap \exists$  PO. River = CityAtRiver.

**German city** — needs the **composition table**. For any country  $c$  with Hamburg PP  $c$ :

- Alster PO Hamburg PP  $c \Rightarrow$  Alster  $\{PO, PP\} c$  comp(PO, PP) =  $\{PO, PP\}$
- Alster overlaps no country ( $\forall PO. \neg \text{Country}$ )  $\Rightarrow$  Alster PP  $c$
- Alster's only country-superpart is Germany  $\Rightarrow c = \text{Germany}$

So Hamburg *is* a German city — and every hidden fact reduces to **concept satisfiability**:

$C \sqsubseteq D$  iff  $C \sqcap \neg D$  unsat.

## $\mathcal{ALCI}_{\text{RCC5}}$ : the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$  = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced **globally**;
- EQ = identity (*strong* semantics).

So one can say things like

$\forall \text{PP}. \neg \text{Lake} \quad \exists \text{DR}. \text{Reserve}$

“no part is a lake”, “disjoint from some reserve”.

**Why this is different**

Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can **quantify over them**. None of the known concrete-domain decidability results apply.

# Why the standard toolbox dies on contact

Every pair of elements carries a relation  $\Rightarrow$  a model is a **complete labelled graph**, never a tree.

## no tree models

Blocking/unravelling —  
the workhorse of DL  
decidability — has no tree  
to work on.

## no finite models

$\exists P P I . T \sqcap \forall P P I . \exists P P I . T$   
forces an infinite ascending  
part-of tower.

## nondeterministic table

Most cells allow several  
outcomes — composition  
*constrains* but rarely  
*determines*.

**Creating one new element forces relations to every existing element** — and a dormant  $\forall$   
anywhere may wake up because of it.



## The landscape: open exactly at the top

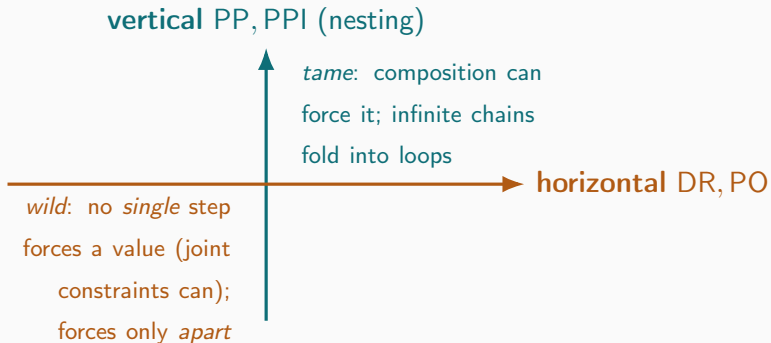
| Logic                                                 | Relations                   | Concept satisfiability           |
|-------------------------------------------------------|-----------------------------|----------------------------------|
| $\mathcal{ALCI}_{\text{RCC1}}$                        | {SR}                        | decidable ( $\equiv$ S5)         |
| $\mathcal{ALCI}_{\text{RCC2}}$                        | {DR, O}                     | decidable                        |
| $\mathcal{ALCI}_{\text{RCC3}}$                        | {DR, ONE, EQ}               | decidable (via two-variable FO!) |
| $\mathcal{ALCI}_{\text{RCC5}}, \forall\text{PO-free}$ | RCC5 w/o $\forall\text{PO}$ | <b>decidable</b> (this project)  |
| $\mathcal{ALCI}_{\text{RCC5}}$                        | all five                    | <b>open</b> ; PSpace-hard        |
| $\mathcal{ALCI}_{\text{RCC8}}$                        | eight RCC8 rel.             | <b>open</b> ; ExpTime-hard       |

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

*“Perhaps the most interesting candidate is  $L_{\text{RCC5}}(\text{RS}) [= \mathcal{ALCI}_{\text{RCC5}}]$  ... to which the reduction exhibited in Section 8 does not apply.”*

Recognized open by the leading experts — for **twenty years**.

# One logic, two completely different axes



- Depth (nesting) is **never the problem** — ordinary modal-logic folding handles it.
- Everything hard lives in **horizontal crowds**: many pairwise-distinct neighbours that no finite summary seems to capture.

## The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly **four** return a single value:

$$\text{comp}(\text{DR}, \text{PPI}) = \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$$

— every one runs through PP/PPI; no single cell yields  $\{\text{PO}\}$ . But the *intersection of several can force PO* ( $\text{comp}(\text{DR}, \text{PO}) \cap \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}\}$ ) — so “shadow” is joint closure, not single-path.

Disjointness is half-deterministic.

**downward:**  $x \text{ DR } y \Rightarrow$  every part of  $x$  is DR from every part of  $y$

**upward:** a superpart of  $x$  may be DR, PO, or PPI to  $y$  — **free**

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for<sub>10/42</sub> free. What resists: PO, and the upward direction.

# Shadows, live width, and the one property that matters

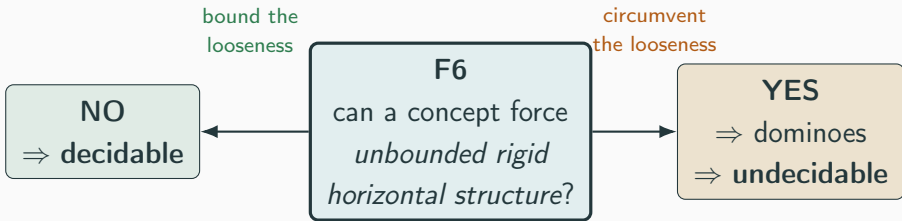
- An edge is a **shadow** if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The **live width** of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

## Property F6 (the keystone of everything)

Is the live width of models bounded by a computable function of the input concept?

If yes  $\Rightarrow$  finite certificate space  $\Rightarrow$  **satisfiability is decidable**.

## The knife's edge: one fact, two futures



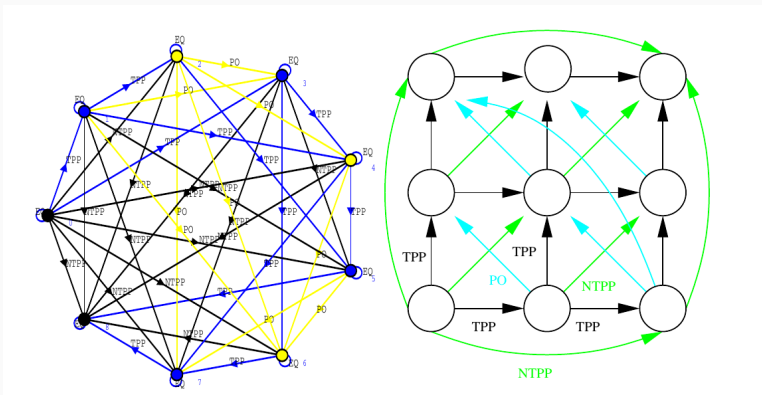
The *same* looseness of the composition table blocks **both** proofs. That is *why* neither direction has been settled in twenty years — it is one question seen from two sides.

## The standard weapon needs a rigid grid

- Classic undecidability proofs encode **domino tilings** of  $\mathbb{N} \times \mathbb{N}$ : build a grid, force tile constraints, reduce the halting problem.
- The crux is **rigidity**: going *east then north* must land on *exactly the same cell* as *north then east* ( $R_X \circ R_Y = R_Y \circ R_X$ , the “coincidence condition”).
- Every known route into  $\mathcal{ALCI}_{\text{RCC}}$  needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals. . .

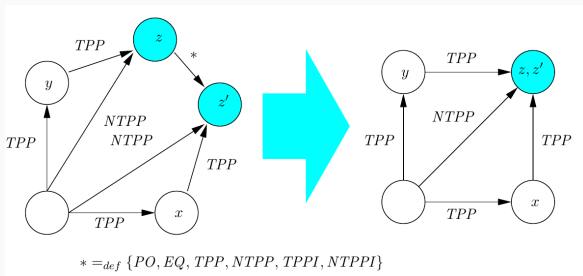
Question (2003 and today): can the composition table itself supply the rigidity?

## 2003: the grid models *exist*...



An RCC8 network isomorphic to an  $n \times n$  grid (report7, Fig. 10, 2002/2003) — the construction *found by a Lisp enumerator* and drawn with CLIM (MCL), not by hand. The models are *out there*.

...but no concept can *force* them: the coincidence obstruction



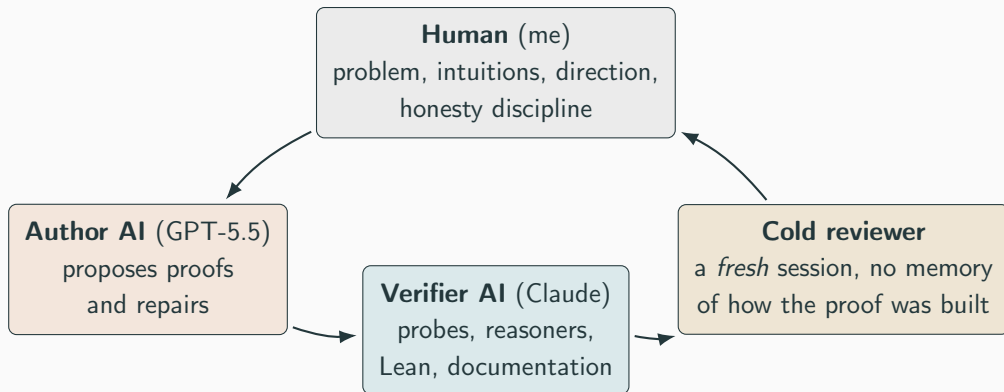
report7, Fig. 11: successors  $z$  (east-north) and  $z'$  (north-east) *should* be one corner — the table *alone* allows **six** relations, forcing neither merge nor split. You *can* force the merge (clash out the five non-EQ options, EQ remains) — but that's the trap: those markers collapse *every* corner at the next level (SAT at level 1, UNSAT at 2; RCC5 *and* RCC8). Neither gives a grid: loose  $\Rightarrow$  no join, forced  $\Rightarrow$  total collapse.

*“Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term.”*

*(report7, 2003)*



# The cast, and the loop



Propose → verify → **attack** → repair. Round after round, March–July 2026.

# The campaign at a glance

- ~4 months, ~30 repair rounds
- 20 adversarial reviews — a defect or overclaim found in *all but two*
- 94 self-contained verification probes (each re-derives the algebra from set semantics)
- 5 Lean files, ~51,000 lines, zero sorry
- 2 working reasoners, cross-validated on 911 concepts

## The single most telling number

Across *all* rounds and reviews, no accepted-but-unsatisfiable concept was ever found. Every defect was a gap in an *argument* — never a counterexample to the intended theorem.

That is why the label is *strongly supported* — and why it is still *not certified*.

## Route 1: tableaux with blocking — broken by totality

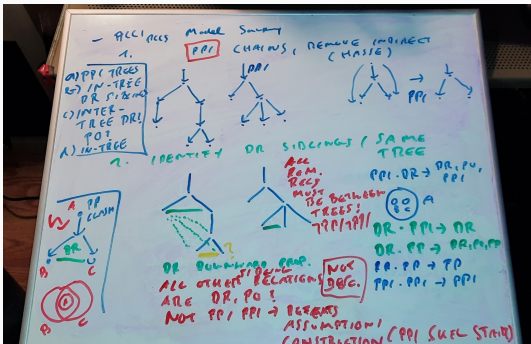
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges **into already-built nodes**: from  $x \text{ PP } y$ ,  $y \text{ DR } z$  the table forces  $x \text{ DR } z$ .
- Diagnostic:  $C_{\text{force}} = \exists \text{PP}.(\exists \text{DR}.A) \sqcap \forall \text{DR}.\neg A$  is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant  $\forall$ 's reawaken blocked nodes; no fixed-arity local summary (types, triangles, quadruples, profiles, ...) is ever enough.

**Lesson:** in a total-relation logic, “local” does not exist.

What may be reused depends on the whole graph. (That is F6 in disguise.)

## Route 2: the split-forest idea (a whiteboard, April 2026)



- **Present** the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are **horizontal**; complete them by the RCC5 *patchwork property*.
- The one component **every** cold review found sound — though the certified route keeps only its *shape*, not this machinery: no patchwork appeal, no quotient.

The human contribution: this decomposition. The models made it precise.

## Route 3: tree automata — “acceptance without validity”

Package the search as a parity tree automaton (emptiness is decidable)...

- ...but the input tree **omits** the relations between fresh witnesses and the sources of  $\forall$ -constraints.
- An *alternating* automaton explores in independent copies — and copies **guess the missing relations independently**.
- A cold review built the killer: two copies privately guess  $\{PP\}$  and  $\{DR\}$  for the same pair;  $\text{comp}(PP, PP) \cap \text{comp}(PP, DR) = \{PP\} \cap \{DR\} = \emptyset$  — every copy accepts, **no model exists**.

So the automaton was **abandoned**, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

# Machine-checking: what the Lean kernel buys

- **Lean 4**: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 **core only** — no libraries; every object built from first principles; tiny trusted base.
- “**zero sorry**” = no admitted gaps anywhere.

## Why we moved to Lean

Most reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and started finding only *interface* gaps — which were then closed.

# What exactly is certified (zero sorry)

| Result                                        | Meaning                                            | File             |
|-----------------------------------------------|----------------------------------------------------|------------------|
| <b>Soundness pipeline</b>                     | valid certificate $\Rightarrow$ real RCC5 model    | Round19Transport |
| <b>Abstraction faithful</b>                   | satisfiable $\Leftrightarrow$ Hintikka-realizable  | Round19Transport |
| <b>RCC5 normal form</b> (both dir.)           | network $\Leftrightarrow$ order + disjointness     | RCC5NormalForm   |
| <b>Forcing reduction</b>                      | F6 fails $\Leftrightarrow$ a concept forces crowds | ForcingReduction |
| <b>Dovetail schema</b>                        | qualitative F6 $\Rightarrow$ decidable             | SemiDecidability |
| $\forall$ PO-free <b>decided</b> (2026-08-29) | <i>a decision procedure</i> , raw input            | P0FreeLift       |
| <b>Abstract = concrete</b>                    | composition table $\Leftrightarrow$ real sets      | P0FreeLift       |

What is **deliberately not** in the kernel: the open completeness premise itself (F6). It is *stated*, never axiomatized — the honest boundary between what is proved and what is conjectured.

# The main theorem (conditional, soundness certified)

## Conditional decidability

If F6 holds — live width bounded by a computable function of the concept — then  $\mathcal{ALCI}_{\text{RCC5}}$  concept satisfiability is **decidable**.

- The **soundness half** is kernel-checked end to end: certificate syntax  $\rightarrow$  closure conditions  $\rightarrow$  proper RCC5 frame  $\rightarrow$  truth lemma  $\rightarrow$  *actual model of the input concept*.
- The certificate checker is a real **executable Boolean function** — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Everything that remains open is the **antecedent** — F6.



# Unconditional win #1: a decidable fragment — now *certified*

**Theorem ( $\forall$ PO-free fragment), machine-checked 2026-08-29**

Concepts with **no  $\forall$ PO subformula** are **decidable**. In Lean, with fragment membership the *only* hypothesis: `decidableFSat (raw input)` `decidableSat_cone (NNF)` `decidableSetSat (real sets)`

- Still allowed:  $\exists$ PO,  $\forall$ DR,  $\exists$ DR and **all** part-of modalities — an expressive spatial fragment.
- On **raw** input the condition is *polarity-sensitive*: no  $\forall$ PO positively, no  $\exists$ PO negatively.
- Reached **three** times by three routes; the third (GPT-5.6 Sol's *cone scheme*) is the one that certified.
- **Decidable  $\neq$  runnable** — doubly exponential.

## Why *that* fragment — the one-sentence reason

### The residual default

When composition leaves a pair's relation open, you can always settle it as **PO** — *unless some formula can object to a PO edge*. The only formula that can is  $\forall \text{PO}.D$ .

- So in the fragment the **lazy arrangement** — declare only what a birth forces, default every remaining pair to PO — is **always** a model.
- Hence **nothing ever has to be coordinated**: every relation between two different children is the residual, and the residual is free. That is why the construction needs no joint realizability.
- Put  $\forall \text{PO}$  back and you do not lose one case of one induction — you lose **the default itself**. That is the whole boundary.

## Is the fragment new? What the literature actually says

Lutz & Wolter 2006 — the one place anyone gestured at this

They leave  $L_{RCC5}(RS)$  — *exactly* this logic — explicitly **open**, and ask for “a decidable and still useful variant”. Among the routes they float: “giving up some of the relations. . . dropping PO still results in a useful formalism”.

- But that sentence defends **usefulness**, not decidability — no proof, no sketch, no conjecture. The work they cite in support has a *single* inclusion modality.
- Our restriction is **weaker**: PO stays in the language.  $\exists PO$  is kept; only  $\forall PO$  goes.
- Wessel’s own 2003 report did canvass restricted fragments — but strictly stronger ones, all working by restoring the **finite model property**, which this fragment provably lacks.

New **so far as we can determine** — deliberately weaker than a priority claim.

## Unconditional win #2: the RCC5 normal form

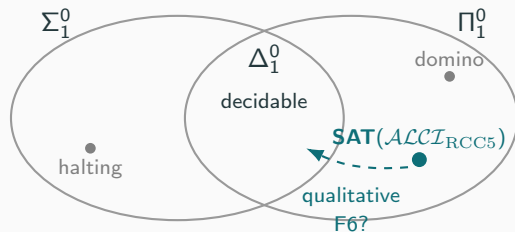
**Theorem (both directions kernel-checked, arbitrary domains)**

A strong-EQ RCC5 network is composition-closed *iff* it is an **ordered-disjoint structure**:

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, **downward-closed** disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited “keeps-apart” relation + overlap as the leftover. The “ $\Rightarrow$ ” is a 150-line kernel proof (propext only); the “ $\Leftarrow$ ” is GPT-5.6 Pro’s **canonical set representation**, also kernel-checked for arbitrary domains — the structural fact both major proof routes silently relied on, now a certified biconditional.

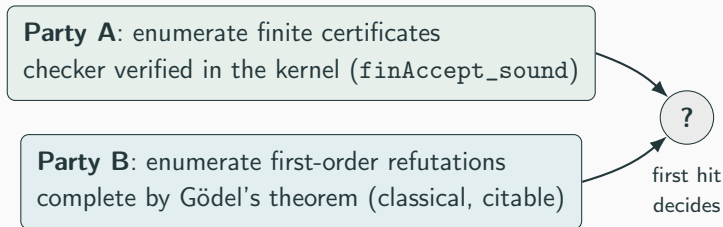
## A late surprise: the problem's exact address is $\Pi_1^0$



- The whole semantics is **finitely first-order axiomatizable**; by Gödel completeness, *unsatisfiability* is enumerable.
- So SAT is **co-r.e.** ( $\Pi_1^0$ ) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Nobody in the project — or the reviews — had noticed for four months. Machine-cross-checked; the reduction is in Lean.

## Consequence: half of the keystone was never needed

A  $\Pi_1^0$  problem is decidable *iff* SAT is also enumerable. So run **two search parties** and wait for the first hit:



- Needed now: only **qualitative** F6 — *some* finite certificate per satisfiable concept, **no computable bound**.
- The bound the reviewers kept demanding (and that kept breaking) comes back **for free, afterwards** — a theorem of the schema.

“Isn’t this just first-order logic?” — yes, and it helps

$\mathcal{ALCI}_{\text{RCC5}}$ -SAT is FO satisfiability of a finite theory. Does it land in a **decidable fragment** of FO?

| Fragment                   | Fate | Why                                                             |
|----------------------------|------|-----------------------------------------------------------------|
| two-variable $\text{FO}^2$ | ×    | composition is ternary (RCC3 <i>did</i> fall this way in 2003!) |
| prefix classes             | ×    | unbounded $\forall\exists$ alternation, FMP                     |
| guarded / clique-guarded   | ×    | <b>totality</b> is the one unguardable axiom                    |
| unary/guarded negation     | ×    | finite model property; totality again                           |
| guarded + transitivity     | ×    | undecidable outside guards                                      |

Every failure traces to the same fact: **every pair is related**, so models are maximally non-local — exactly the obstruction F6 names. The FO view rediscovers the project’s wall from classical model theory.

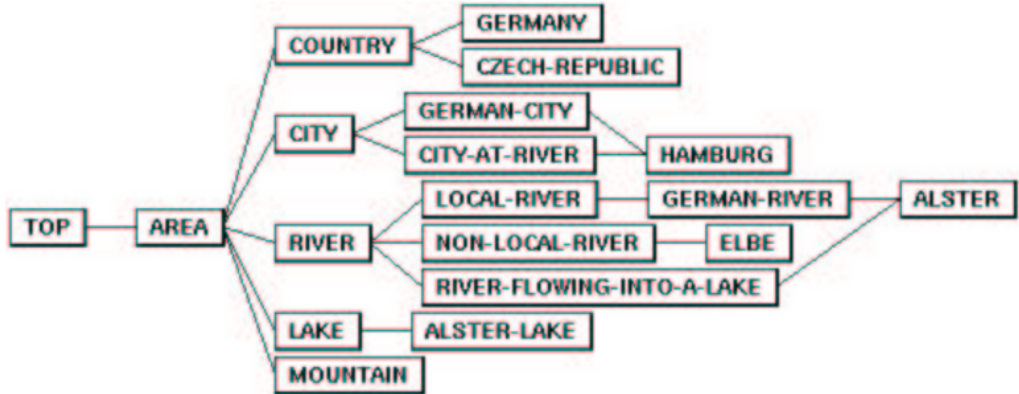
## A working prototype reasoner (validated, not proved)

- The **cover-tree tableau** ( $\sim 350$  lines of Python): exact-occurrence blocking on the split-forest foundation.
- Cross-validated against an independent oracle on **911 concepts — zero mismatches**; 775/775 structured decomposition tests; never once contradicted in the whole campaign.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

A running algorithm and a kernel-checked proof, **pointing the same way** — the strongest evidence short of a certificate.

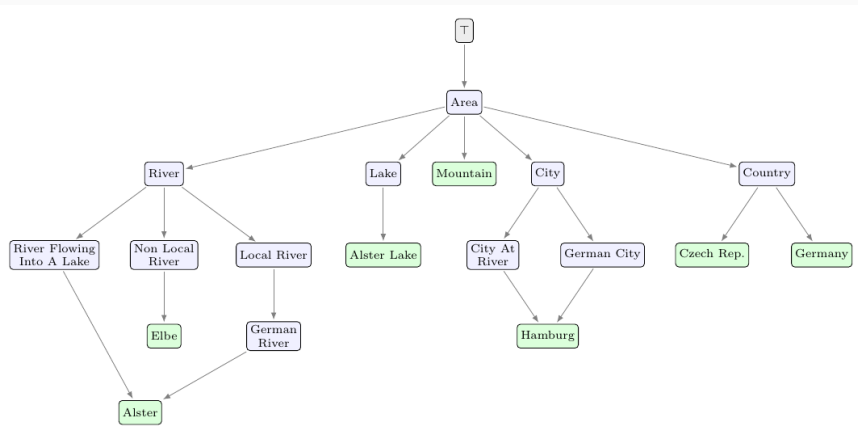


## 2003: a prototype reasoner



The GIS concept taxonomy in report7 (Fig. 6), 2003 — itself *computed by a prototype reasoner* of the era, not drawn by hand.

## 2026, the cover-tree reasoner: all 21/21 — edge for edge



The 2026 cover-tree tableau reproduces it exactly — **two reasoners, 23 years apart, identical result**. (Green = individuals; *Hamburg* and *Alster* each have **two** parents — a DAG, not a tree.)<sub>33/42</sub>

# What the models did — and what they could not

## Demonstrated

- months-long rigorous formal development,  $\sim 30$  rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a  $\sim 51,000$ -line kernel-checked Lean development
- working, validated software

## Not demonstrated

- the **decisive novel idea** — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. **What carried the project was the loop, not any single model.**

# The failure mode worth naming: productive-looking thrashing

Three weeks. Four successive repairs to one architecture. Each locally reasonable, each probed, each refuted by an exact finite countermodel — and it proposed the next one.

- It never concluded the *class* was wrong — though its own design document already recorded this as the *fifth* occurrence of the same shape.
- Not hallucination, not local error: *every step was correct and checked*. The context makes the framing feel settled because it was never questioned there.
- What broke it came from outside: a halt, and a *different model* — no history in the design — asked for a different architecture. That answer became the certified route.

Cold review, aimed at *design* rather than proof. Knowing *when* to call it was the human's —  
*we cannot automate that.*

## The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** Every load-bearing claim was cross-checked mechanically: 94 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, **not the model's confidence**.
3. **Label honestly.** *Strongly supported*  $\neq$  *certified*. Overclaiming happened twice; cold review caught both; the claims were withdrawn — **on the record**.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

## If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an **existence proof** of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should **know** these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it is now on arXiv, **in the open**, with the full audit trail.

*The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.*

# The results, by verification level

| Level          | Results                                                                                                                                                                                                                     | What it means                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Lean-certified | $\forall$ PO-free <b>decidability</b> (raw input, NNF, concrete sets); abstract = concrete sat.; soundness pipeline; abstraction faithfulness; RCC5 normal form (both dir.); $F6 \Rightarrow$ decidable; $\Pi_1^0$ dovetail | kernel-checked, with a stated axiom footprint |
| Theorem-level  | the <i>two arguments</i> for the fragment ( $K(C_0)$ bound, extraction); identity-selector characterization; $W2'$ folds into F6                                                                                            | proof in a work package, often with probes    |
| Empirical      | cover-tree tableau; 911-concept cross-validation; model verifier                                                                                                                                                            | test evidence, not a decision theorem         |
| Open           | static F6; completeness of the certificate scheme; automatic-cover property; full decidability                                                                                                                              | no proof claimed                              |

Four levels kept rigorously distinct — conflating them is the easiest way to misread the project.

### One well-posed question

can a concept force unbounded  
*identity-selector* structure?

- NO  $\Rightarrow$  decidable, YES  $\Rightarrow$  the undecidability route opens — certified machinery waits on **both** sides.
- One **sub-problem off the pile**: drop  $\forall$ PO and it is decided, in the kernel — three cold reviews, no counterexample.
- Reproducible: 5 Lean files ( $\sim$ 51k lines, 0 sorry), 94 probes, 2 reasoners, full log.

**Open since 2003. Still standing — but one wall has come down.**

[github.com/lambdamikel/alcircc5](https://github.com/lambdamikel/alcircc5) — thank you!



## Backup: the full RCC5 composition table

| comp        | $DR(a, b)$ | $PO(a, b)$  | $EQ(a, b)$ | $PPI(a, b)$     | $PP(a, b)$ |
|-------------|------------|-------------|------------|-----------------|------------|
| $DR(b, c)$  | *          | DR, PO, PPI | DR         | DR, PO, PPI     | DR         |
| $PO(b, c)$  | DR, PO, PP | *           | PO         | PO, PPI         | DR, PO, PP |
| $EQ(b, c)$  | DR         | PO          | EQ         | PPI             | PP         |
| $PP(b, c)$  | DR, PO, PP | PO, PP      | PP         | PO, EQ, PP, PPI | PP         |
| $PPI(b, c)$ | DR         | DR, PO, PPI | PPI        | PPI             | *          |

Row =  $S(b, c)$ , column =  $R(a, b)$ , entry = possibilities for  $(a, c)$ ; \* = all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

## Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
  - a *core network* — the non-repeating part of the model;
  - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
  - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

## Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all **collapse** to finite catalogues — machine-checked.
- The sharp obstruction:  $n$  pairs  $(L_i, U_i)$ , each kept apart by a **private** selector  $A_i$  and by nothing else — the selector incidence matrix is the  $n \times n$  *identity*.
- A finite catalogue can hold only boundedly many private selectors, so: **F6 fails iff some concept forces such minors for every  $n$ .**
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use  $\mathcal{ALCI}_{\text{RCC}}$ -*forceability*, not RCC5 consistency alone. The uniformization side-condition (W2') provably folds into F6.